Shahriar Iqbal Zame

Ph.D. Researcher (TSE, TUM)
M.Sc. in Transportation Systems, TUM
Johannes B. Ortner Foundation Prize 2025, TUM
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Summary

I am a Research Associate and PhD candidate at the Technical University of Munich (TUM), Chair of Transportation Systems Engineering. My research on the DFG-funded project "Shaping E-Roads Efficiency for Freight Automation" (SEE-FAr) uses agent-based simulation to advance the electrification and automation of freight transport, exploring charging solutions that improve sustainability and economic efficiency in long-haul logistics. My expertise spans agent-based modeling, optimization, choice modelling and large-scale simulation, with experience in MATSim, high-performance computing, and programming in Python, Java and R.

Education

2025 – Present
Research Associate & Ph.D. Candidate,
Technical University of Munich (TUM), Munich, Germany
Project: *Shaping E-roads Efficiency for Freight Automation (SEE-FAr)*
A joint initiative of TUM's TSE and BOKU's IVe
Supervisors: Prof. Dr. Constantinos Antoniou, Prof. Dr. Yusak Susilo
Member, EIT Urban Mobility Doctoral Training Network (DTN)

2021 – 2024
M.Sc. in Transportation Systems,
Technical University of Munich (TUM), Munich, Germany
CGPA 1.6/4.00 (best 1.0 out of 4.0)
Thesis: *UAM siting considering capacity and land use in Munich* (*Graded 1.0*)
Supervisors: Prof. Dr. Constantinos Antoniou, M.Sc. Hao Wu
Johannes B. Ortner Foundation Prize — outstanding early-career research; €1,500.

2014 – 2017
B.Sc. in Civil and Environmental Engineering
Islamic University of Technology, Bangladesh
CGPA 3.37/4.00 (best 4.0 out of 4.0)
Thesis: *Comparing Comfort in Public Transports in Dhaka, Bangladesh*

Experience

2017 – 2022
Research Engineer, **Icon Engineering Services Ltd.**, Bangladesh
Projects: Dhaka Metro Rail, First Dhaka Elevated Expressway, Shahjalal Airport Terminal 3
Responsibilities: Structural Quality Control of 750+ foundations, Managed **Commercial Website** Development, Published **Corporate brochure** & **Multimedia Content**.

Research Publications

4 Who benefits from Urban Air Mobility? Spatial and income equity in multi-phased vertiport deployment.
Shahriar Iqbal Zame, Tao Guo, Hao Wu, Constantinos Antoniou
Journal of Urban Mobility, Volume 10, 2026, 100304.

3 Comparative analysis of interpretable data-driven and analytical car-following models on highways: Robustness, sensitivity and transferability.
Filippos Adamidis, **Shahriar Iqbal Zame**, Constantinos Antoniou
IET Intelligent Transport Systems, Volume 20, Issue 1, 2026, e70240.

2 A sustainable multi-objective framework for multi-phased, capacitated vertiport siting with land use integration.
Hao Wu, **Shahriar Iqbal Zame**, Tao Guo, Qing-Long Lu, Constantinos Antoniou
Communications in Transportation Research, Volume 5, 2025.

1 Data-driven vertiport siting: A comparative analysis of clustering methods for Urban Air Mobility.
Tao Guo, Hao Wu, **Shahriar Iqbal Zame**, Constantinos Antoniou
Journal of Urban Mobility, Volume 7, 2025

Academic Service & Training

Peer Review

2026 *Journal of Urban Mobility* (Elsevier) — 2 manuscripts reviewed
2026 28th Annual Conference of the EURO Working Group on Transportation (EWGT 2026), Kraków, Poland — 5 manuscripts reviewed

Workshop Organization

Jul 2026 Organizing team, 1st SEE-FAr Workshop: *Recent Advances in Electric and Automated Road Freight Transport*, TUM

Talks

Jul 2026 *Freight OD Estimation from Fleet GPS and Traffic Counts: What Bavaria Validated and Austria Taught Us* — 1st SEE-FAr Workshop, TUM

Training

Jan 2026 *Discrete Choice Analysis: Predicting Individual Behavior and Market Demand*, TRANSP-OR, EPFL, Switzerland
Random utility models, nested and mixed logit, hybrid choice models, Bayesian estimation and forecasting; hands-on estimation in BIOGEME.

Skills

Modeling	Agent-based transport simulation (MATSim and extensions), macroscopic 4-step travel demand modeling, multimodal network modeling for urban mobility systems.
Optimization	Multi-objective optimization, routing and scheduling (e.g., TSP, VRP), heuristic and algorithmic approaches for transport and infrastructure planning.
Programming & AI	Java (OOP), Python, R, MATLAB, SQL, Shell scripting, Pandas, NumPy, Scikit-learn, Machine learning, Transfer learning.
Technical Tools	High-performance computing (Linux, Slurm), geospatial toolchains (QGIS, PostGIS, GDAL), scripting and version control (Python, Bash, Git), development environments (Jupyter, VS Code).
Research	Peer-reviewed academic writing, systematic literature reviews, large-scale transport simulations, interdisciplinary research, LaTeX typesetting.

Creative Projects & Volunteering

2023 Bicycling from Kufstein to Trento (300+ km in 3 days)
2017 ***CycleDhaka:*** *Champion of Inter-University Film Fest (Cennovation, IUT)*
2016 *Volunteering: Light of Hope (Children Education)*
2015 *Volunteering: It's Humanity Foundation (Fighting Poverty)*